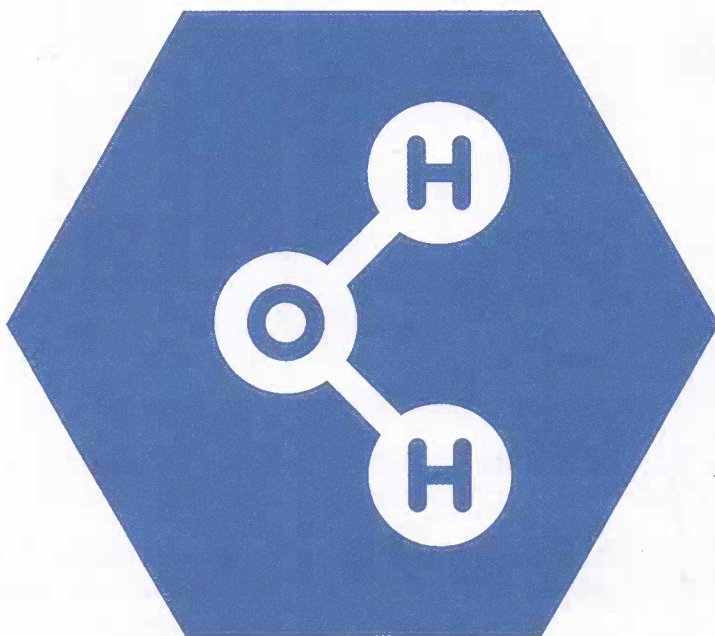




UNSW Global
THE UNIVERSITY OF NEW SOUTH WALES
SYDNEY • AUSTRALIA

PAPER C



2014 ICAS

International Competitions
and Assessments for Schools

SCIENCE

DO NOT OPEN THIS BOOKLET UNTIL INSTRUCTED.

40 QUESTIONS

TIME ALLOWED: 55 MINUTES

STUDENT'S NAME:

Read the instructions on the **ANSWER SHEET** and fill in your **NAME, SCHOOL** and **OTHER INFORMATION**.
Use a 2B or B pencil.
Do **NOT** use a pen.
Rub out any mistakes completely.

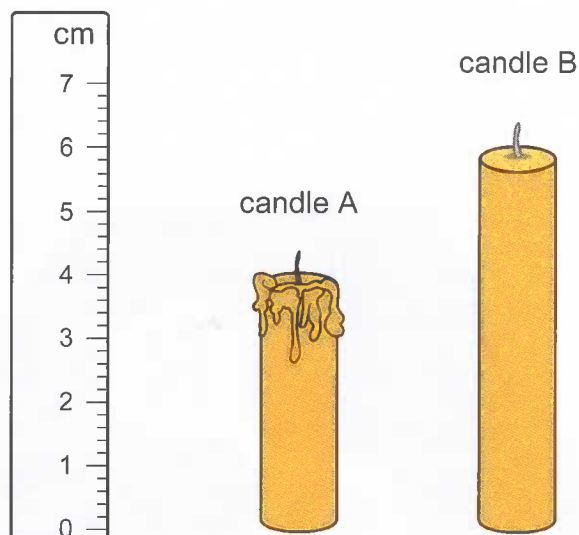
You **MUST** record your answers on the **ANSWER SHEET**.

Mark only **ONE** answer for each question.
Your score will be the number of correct answers.
Marks are **NOT** deducted for incorrect answers.

There are **40 MULTIPLE-CHOICE QUESTIONS** (1–40).
Use the information provided to choose the **BEST** answer from the four possible options.
On your **ANSWER SHEET** fill in the oval that matches your answer.

You may use a calculator and a ruler.

1. Milan had two identical candles. Milan lit candle A and let it burn for one hour. He then blew the flame out. Milan measured the height of the candles with a ruler.



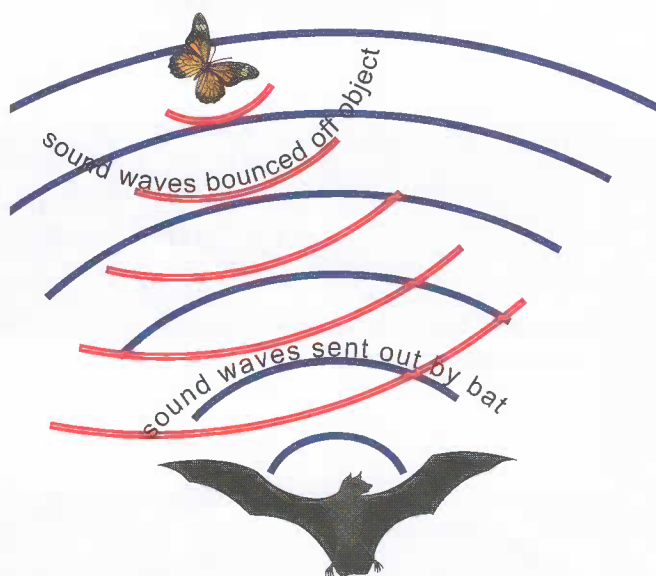
Milan then lit candle B.

How long will it take for candle B to burn completely?

- (A) 2 hours
- (B) 3 hours
- (C) 4 hours
- (D) 6 hours

2. Echoes are produced when sound waves bounce off solid objects. Bats use echoes to determine the distance between them and objects in their surroundings.

A bat sends out a series of high-pitched sounds. The times it took for sound waves to reach some objects and bounce back to the bat are recorded in the table.



Object	Time taken for sound to reach object and bounce back to bat (seconds)
butterfly	0.25
tree	0.30
building	0.35
bird	0.20

Based on the data in the table, which conclusion is correct?

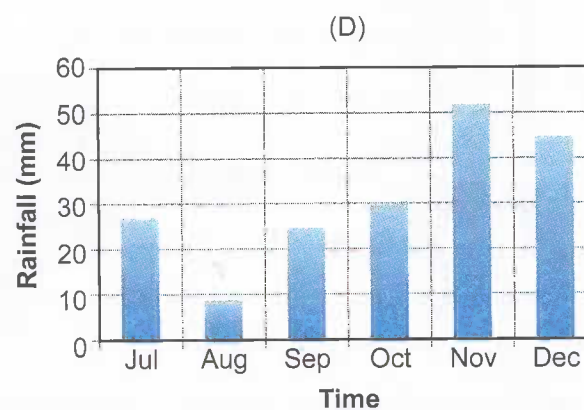
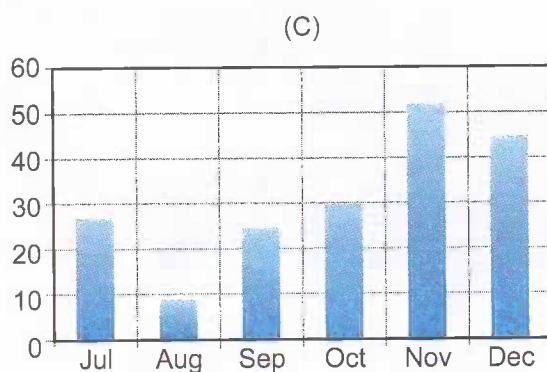
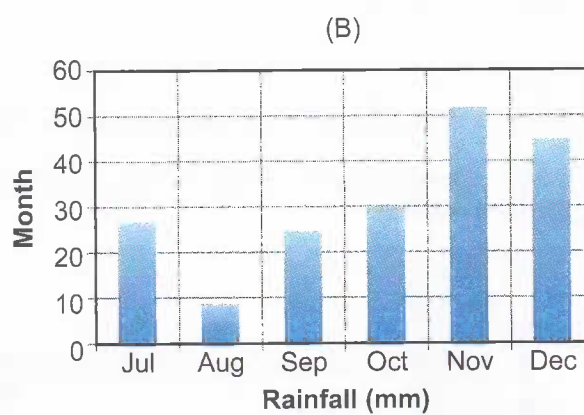
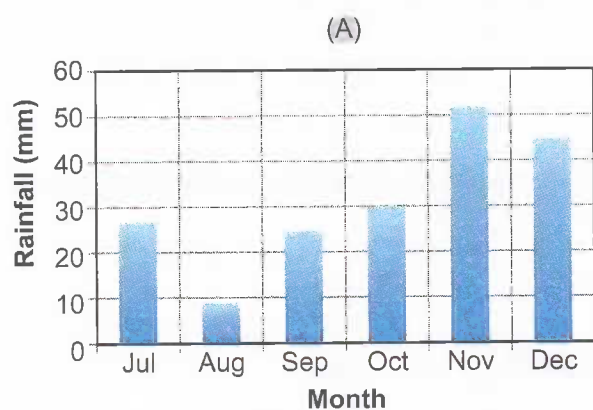
- (A) The bird is flying towards the bat.
- (B) The building is the furthest object from the bat.
- (C) The butterfly is the closest object to the bat.
- (D) The bird is further from the bat than the building.

3. Sue measured the total monthly rainfall in her suburb for 6 months.

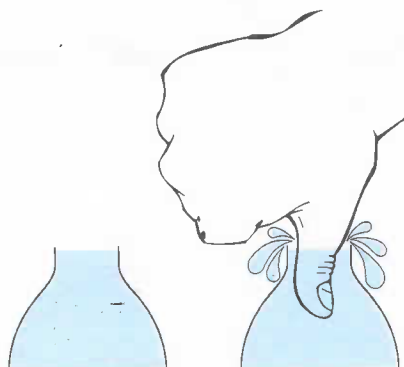
The table shows Sue's data.

Month	Total Rainfall (mm)
July	27
August	9
September	25
October	30
November	52
December	45

Which graph best represents Sue's data?



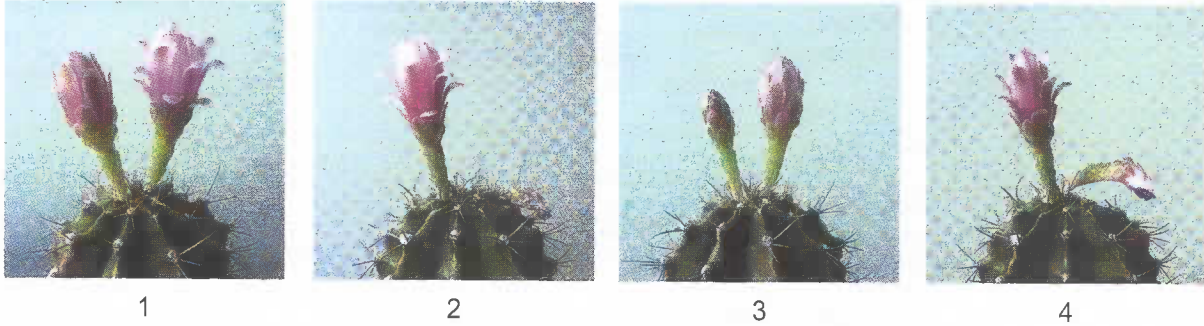
4. In an experiment, the volume of a thumb was measured as shown in the diagram.



Which of the following methods will give the volume of the thumb?

- (A) measure how much water is in the bottle
- (B) measure how much water is left on the thumb
- (C) measure how much water it takes to refill the bottle
- (D) measure how deep the thumb goes into the water

5. The photographs show a plant as it is flowering. The photographs are **NOT** shown in the order in which they were taken.

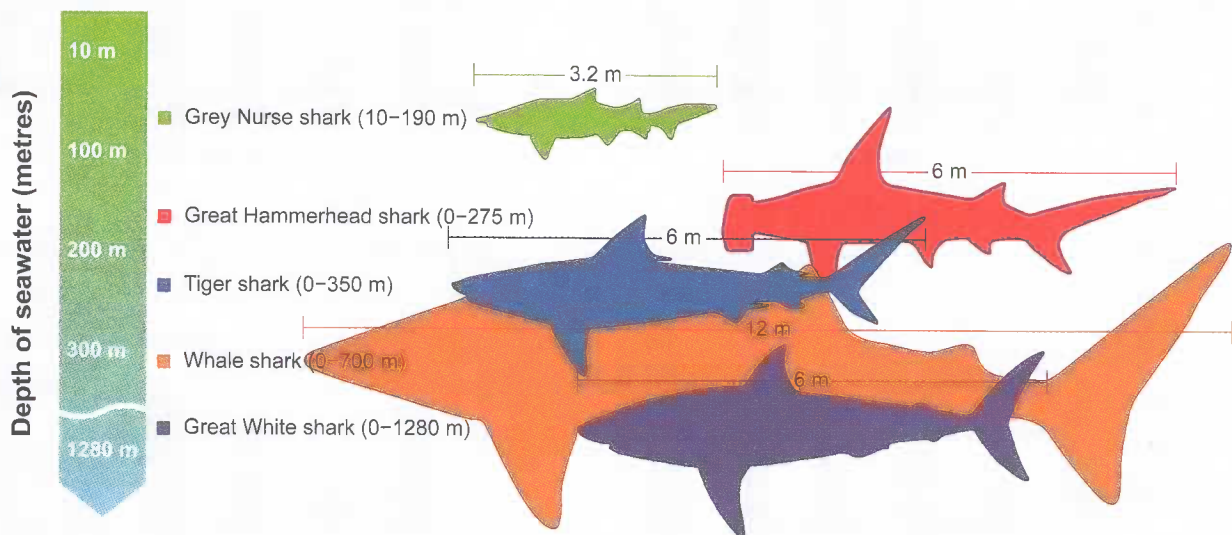


In which order were the photographs taken?

- (A) 1, 3, 2, 4
(B) 2, 3, 4, 1
(C) 3, 1, 4, 2
(D) 4, 2, 3, 1

For questions 6 and 7 use the information below.

This diagram shows the name and size of different adult sharks and the depth of seawater at which each shark is found.



6. Which sharks would be found at depths between 300–750 m?
- (A) Grey Nurse and Tiger shark only
(B) Great Hammerhead and Whale shark only
(C) Grey Nurse, Tiger and Whale shark only
(D) Great White, Tiger and Whale shark only
7. Which shark is **NOT** found at the surface?
- (A) Great White
(B) Whale
(C) Tiger
(D) Grey Nurse

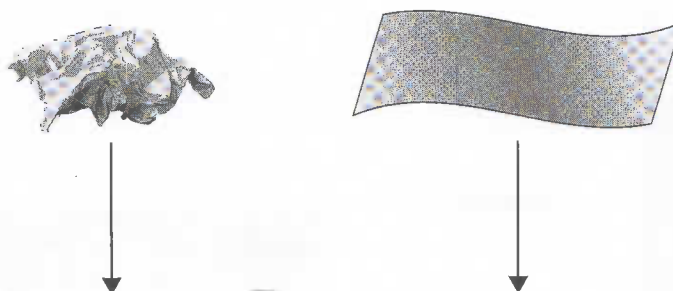
8. The flakes of some breakfast cereals contain iron filings.

Which would be the best way to separate the iron filings from the cereal?

- (A) use a magnet
- (B) use a magnifying glass
- (C) crush the flakes and then use a magnet
- (D) crush the flakes and then use a magnifying glass

9. When an object is dropped from a height, gravity causes it to fall to the ground. Sam took two identical sheets of paper. He crushed one sheet of paper and left the other one flat.

He then dropped them from the same height as shown.



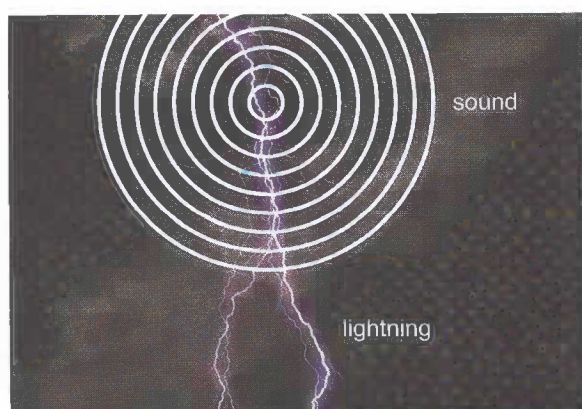
Which sheet of paper will hit the ground first and for what reason?

	Paper that will hit the ground first	Reason
(A)	crushed sheet	It is heavier.
(B)	flat sheet	There is more air pushing it back.
(C)	crushed sheet	There is less air pushing it back.
(D)	both hit the ground at the same time	Both have the same mass.

10. Sound travels at a speed of about 330 metres per second in air. Light travels at 300 000 kilometres per second in air.

You see the lightning before you hear the thunder during a thunderstorm, because light travels faster than sound.

The time difference between seeing the lightning and hearing the thunder lets you calculate how far away you are from the storm.



During a thunderstorm, Lina saw lightning and nine seconds later heard thunder.

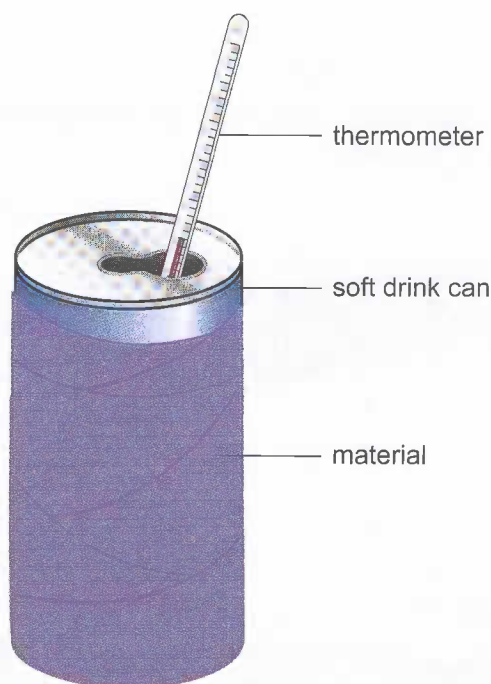
Approximately how far away was the storm?

- (A) 3 km
- (B) 30 km
- (C) 297 km
- (D) 2700 km

For questions 11 and 12 use the information below.

Amy did an experiment to test how well different materials reduce heat loss.

She wrapped three identical empty soft drink cans in a different material. She then put equal amounts of water at 50 °C into each can. After five minutes she measured the water temperature and recorded it in a table.



Material	Water temperature (°C) after five minutes
cotton	39
nylon	44
wool	42

11. Which of the following lists the materials from most effective at reducing heat loss to least effective?

- (A) nylon, wool, cotton (B) nylon, cotton, wool
(C) wool, nylon, cotton (D) wool, cotton, nylon

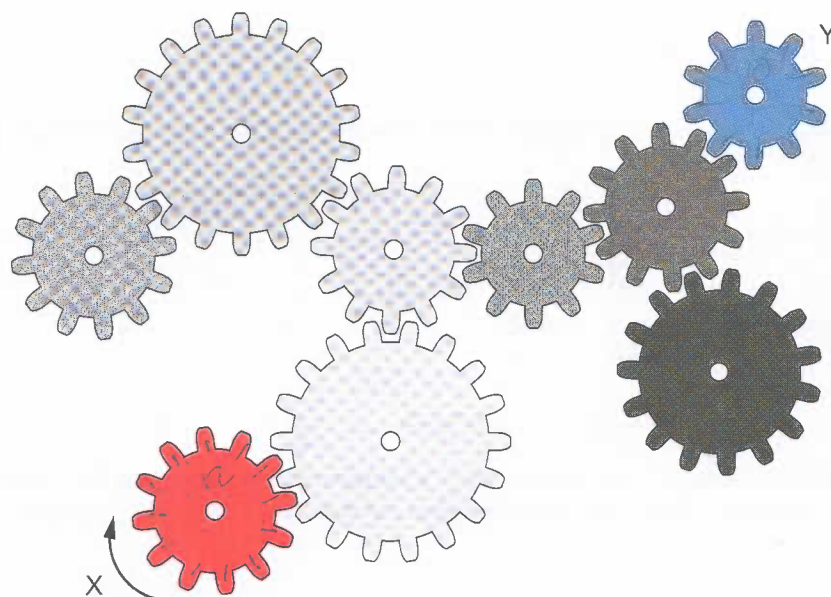
12. Amy wanted to use a control so she could compare her results.

Which of the following would be the best control for this experiment?

	Water amount	Water temperature	Covering materials
(A)	same as before	50 °C	all three together
(B)	same as before	50 °C	no material
(C)	same as before	room temperature	no material
(D)	no water	room temperature	no material

13. When one gear turns another gear, the gear with more teeth will rotate more slowly than the gear with fewer teeth. The two gears will turn in opposite directions.

The diagram represents a number of gears joined together.



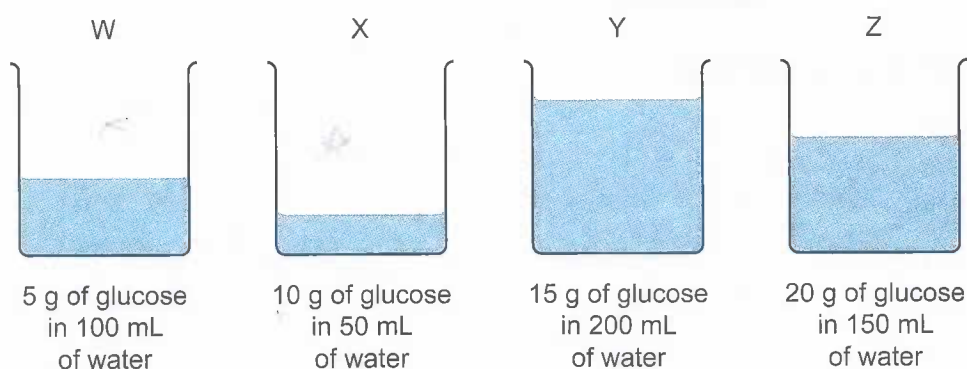
Gear X turns clockwise.

Which option describes the turning of gear Y?

- (A) clockwise and faster than gear X
(B) clockwise and slower than gear X
(C) anticlockwise and faster than gear X
(D) anticlockwise and slower than gear X

14. Glucose is a type of sugar.

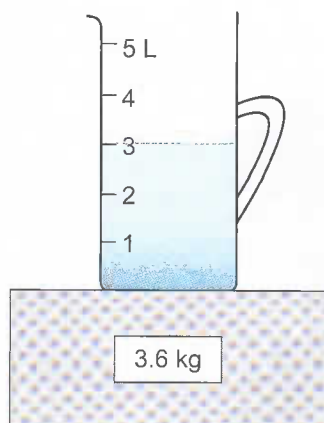
A student dissolved different amounts of glucose in different volumes of water. The water temperature in each container was the same.



Which solution would taste the sweetest?

- (A) W (B) X (C) Y (D) Z

15. Wendy measured the mass of a plastic jug containing water. The mass of one litre (L) of water is one kilogram (kg).

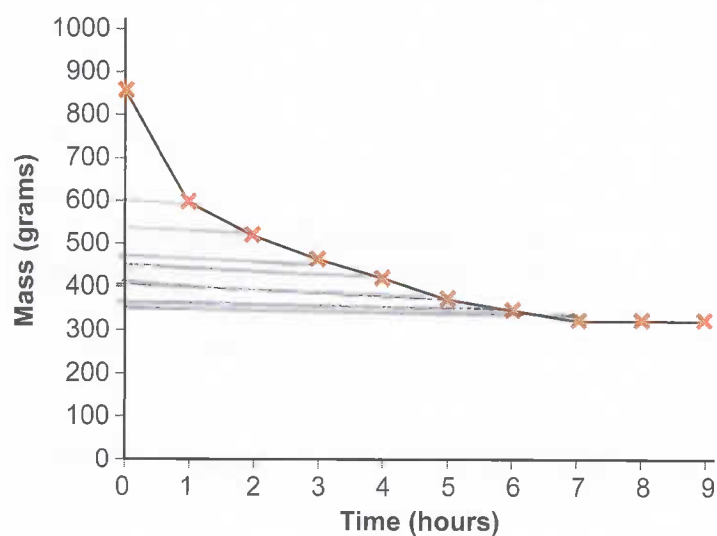


The diagram shows the mass of the jug including the water it contained.

What is the approximate mass of the jug (in kg)?

- (A) 0.6
- (B) 2.0
- (C) 2.6
- (D) 3.0

16. Serena and Bill wanted to find out how long it takes a sweater to dry. They washed a new sweater, weighed it and then hung it out to dry. Every hour they weighed the sweater. The graph shows their results.



How long did the sweater take to dry?

- (A) 6 hours
- (B) 7 hours
- (C) 8 hours
- (D) 9 hours

17. The carbon cycle is the flow of carbon through Earth, including the air and the bodies of plants and animals.

One example of a carbon cycle includes these stages:

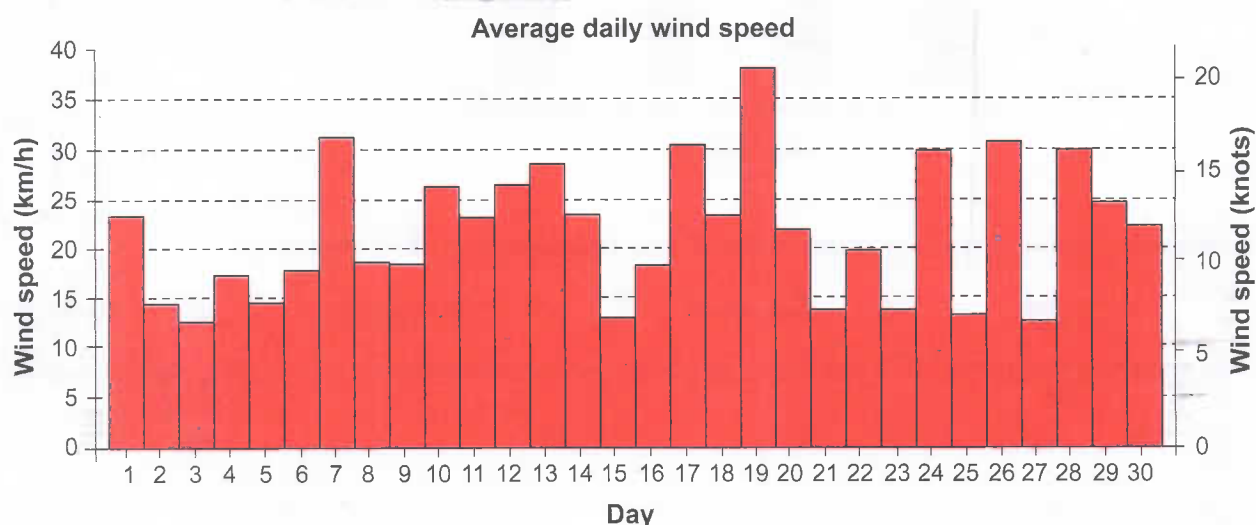
1. Plants absorb carbon dioxide from the air to produce sugar.
2. A pig dies and a crocodile eats its carcass.
3. A pig eats plants.
4. A crocodile dies and decays, releasing carbon dioxide into the air.

What is the correct order for the stages in this example?

- (A) 1, 3, 2, 4 (B) 2, 3, 1, 4 (C) 3, 4, 2, 1 (D) 1, 3, 4, 2

For questions 18 and 19 use the information below.

This graph shows the average daily wind speed in an area over a period of 30 days.



18. What was the highest average daily wind speed, in kilometres per hour (km/h), in the 30-day period?

- (A) 38 (B) 21 (C) 12 (D) 7

19. The wind speeds are given in two different units, kilometres per hour and knots.

Approximately what speed, in kilometres per hour, is equal to one knot?

- (A) 1.5 (B) 1.9 (C) 10 (D) 18

20. The composition of pure sulfuric acid (H_2SO_4) by mass is 2% hydrogen, 65% oxygen and the rest is sulfur.

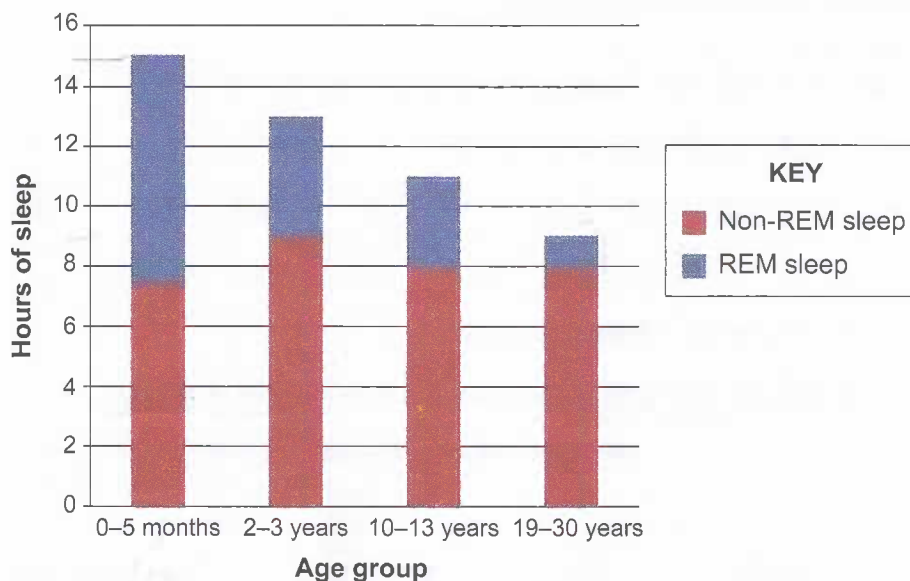
What percentage of the mass of sulfuric acid is made up of sulfur?

- (A) 7%
(B) 14%
(C) 33%
(D) 100%

For questions 21 and 22 use the information below.

Rapid eye movement (REM) sleep is a stage during sleep when the eyes of the sleeping person move rapidly.

The graph represents the hours of REM and non-REM sleep for people in different age groups.



21. Which age group spends the greatest fraction of their total sleep-time in REM sleep?
- (A) 0-5 months (B) 2-3 years (C) 10-13 years (D) 19-30 years
22. How many hours more than the 19-30 year age group does the 0-5 month age group spend sleeping per day?
- (A) 15 (B) 9 (C) 7 (D) 6

23. Matt was walking along the sand when he saw these footprints.



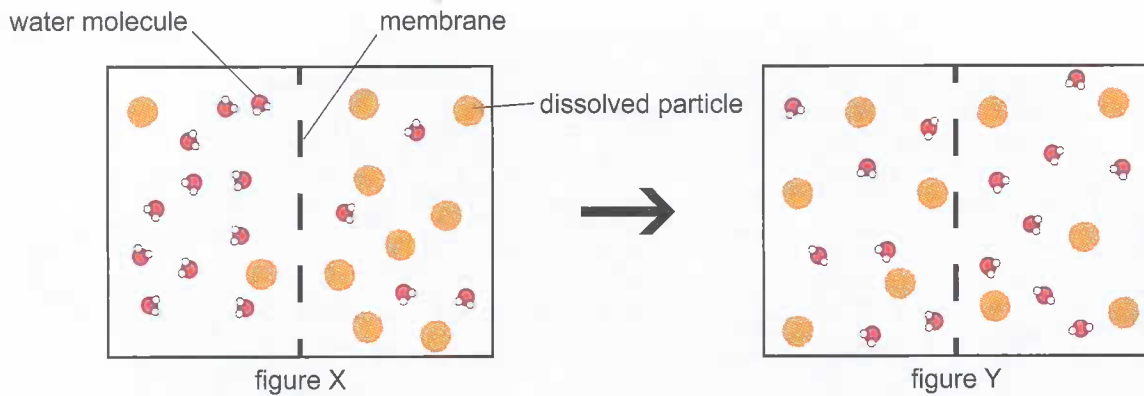
Which option shows the correct order of footprints on the sand, from first footprint to last footprint?

- (A) cat, horse, human, bird (B) bird, human, horse, cat
- (C) horse, human, bird, cat (D) human, bird, horse, cat

24. Diffusion is the movement of dissolved particles from an area of higher concentration towards an area of lower concentration.

Osmosis is the movement of water molecules through a membrane from an area of higher concentration towards an area of lower concentration.

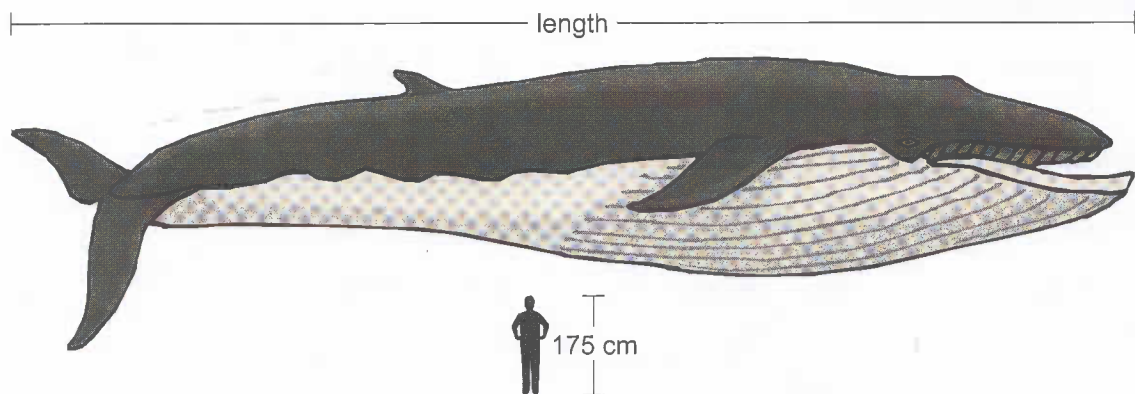
The diagram below shows water molecules and dissolved particles in one container separated by a membrane that allows both these substances to pass through it.



What has happened to the solution in figure X to cause the arrangement of substances shown in figure Y?

- (A) diffusion only
(B) osmosis only
(C) diffusion and osmosis
(D) neither osmosis nor diffusion

25. The Blue Whale, a marine mammal, is the largest animal known to have lived on Earth.



The man and the Blue Whale are drawn to the same scale.

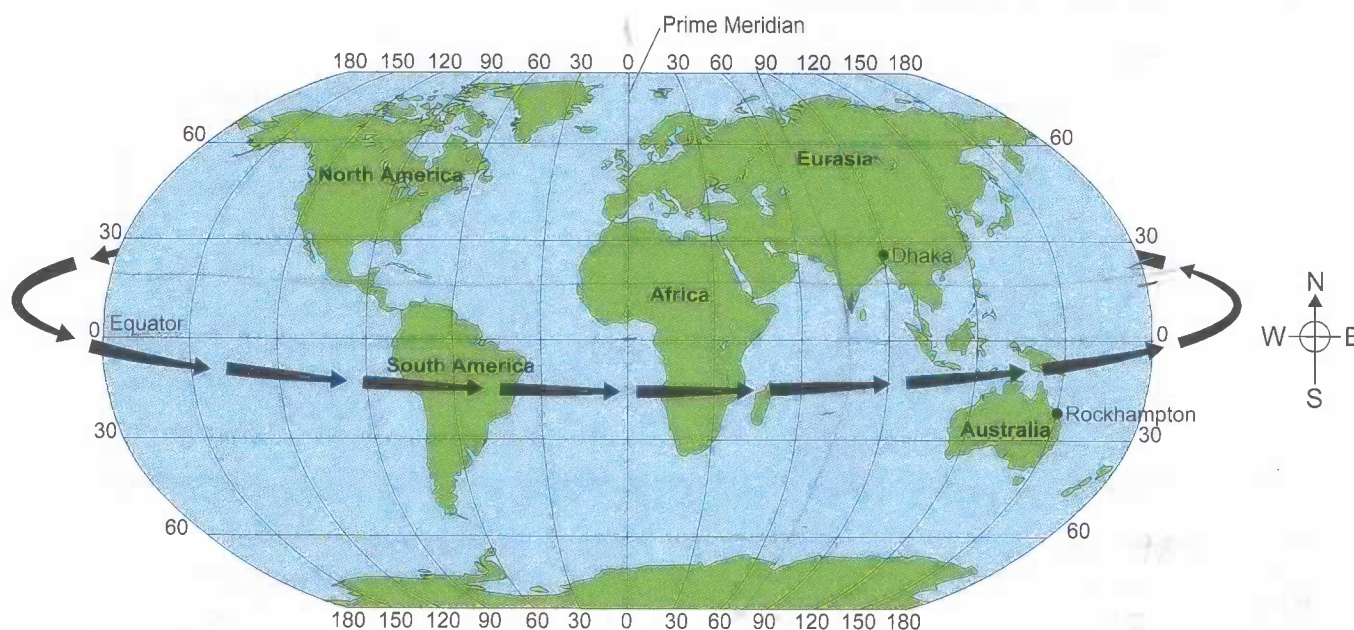
What is the approximate length of the Blue Whale?

- (A) 20 m
(B) 30 m
(C) 40 m
(D) 50 m

For questions 26 and 27 use the information below.

Places on the surface of Earth are given a location using coordinates called latitude and longitude. These coordinates locate places by their distance from the Equator and the Prime Meridian.

One complete revolution of Earth occurs once every 24 hours. The planet is divided into 24 time zones one hour apart.



26. A fossil was found at location 20° N and 75° E.

Where was the fossil found?

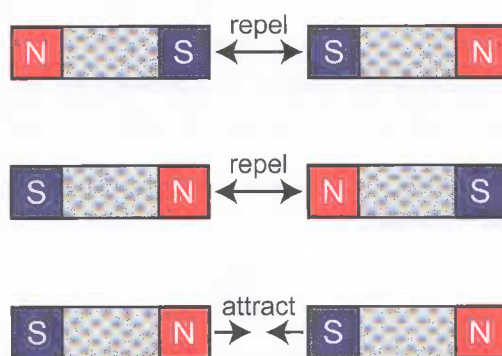
- | | |
|-------------|-------------------|
| (A) Eurasia | (B) Australia |
| (C) Africa | (D) South America |

27. The longitude of Rockhampton is approximately 150° E and the longitude of Dhaka is 90° E.

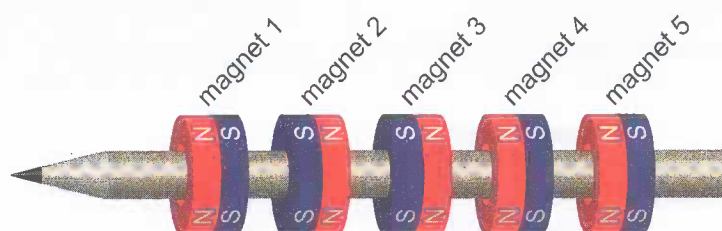
What is the time in Dhaka when it is 6 am in Rockhampton?

- | | | | |
|----------|-----------|-----------|----------|
| (A) 2 am | (B) 10 am | (C) 10 pm | (D) 4 am |
|----------|-----------|-----------|----------|

28. Magnets have a north pole and a south pole. When magnets are placed near one another, they can either attract or repel one another as shown in the diagram.



Frans placed five strong ring magnets onto a pencil as shown in the diagram.

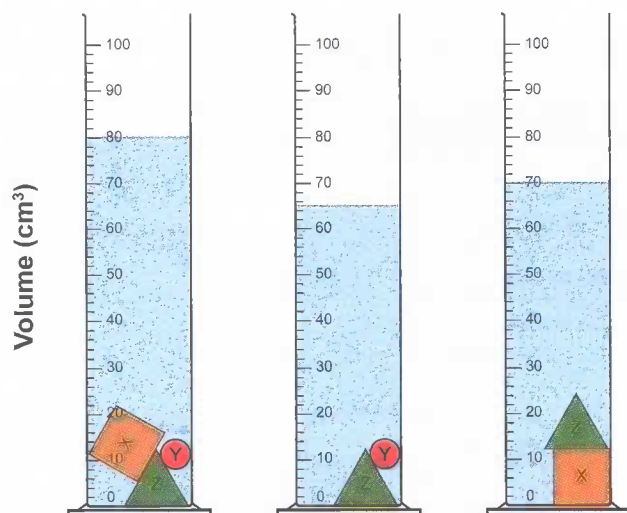


Which option correctly states what will happen to some of the magnets?

	Magnets that will repel each other	Magnets that will attract each other
(A)	magnet 1 and magnet 2	magnet 3 and magnet 4
(B)	magnet 3 and magnet 4	magnet 2 and magnet 3
(C)	magnet 4 and magnet 5	magnet 1 and magnet 2
(D)	magnet 2 and magnet 3	magnet 4 and magnet 5

29. The volume of an object is equal to the volume of water it displaces.

Jason set up an investigation to find the volume of several objects using measuring cylinders and 50 mL of water in each measuring cylinder.



What are the volumes of objects X, Y and Z?

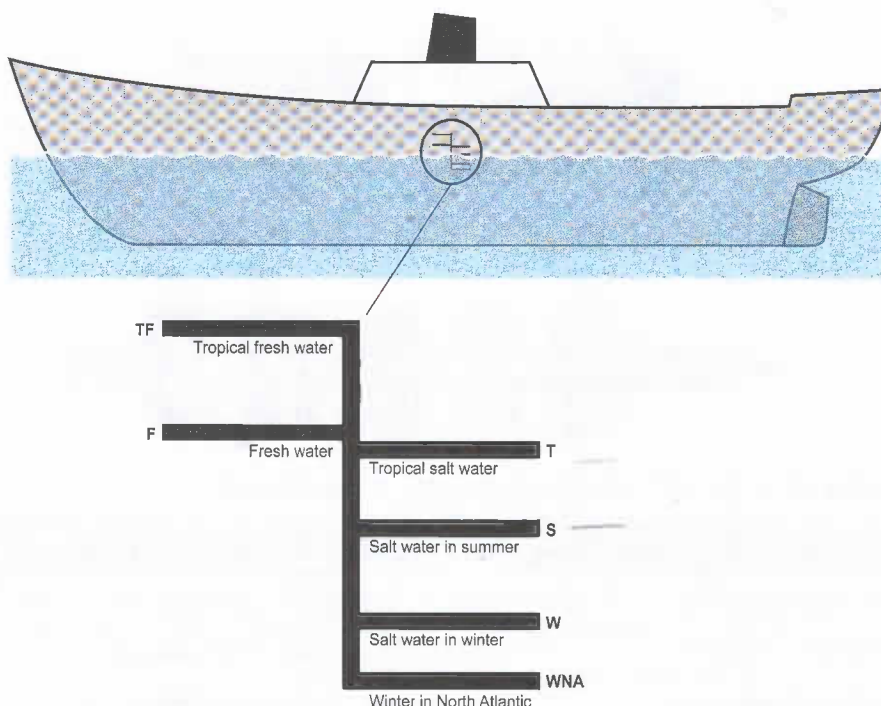
	Volume (cm ³)		
	X	Y	Z
(A)	5	10	55
(B)	10	15	5
(C)	15	5	5
(D)	15	10	5

For questions 30 and 31 use the information below.

Plimsoll lines are markings on the side of a ship. They indicate the limits to which a ship can be safely loaded for journeys around the world.

Objects float when the upward force from the water is greater than the downward force (weight) of the floating object. The upward force provided by the displaced water is called buoyancy.

Warm water provides less buoyancy than cold water so the ship sits lower in the water. Fresh water provides less buoyancy than salty seawater.



30. Of what use are Plimsoll lines on cargo ships that sail around the world?

- (A) so we can tell if the ship is in warm or cold water
- (B) so we can tell if the ship is in fresh or salt water
- (C) to make sure the ship carries the correct cargo
- (D) to make sure the ship always carries a safe amount of cargo

31. A ship is correctly loaded with cargo in a warm water sea port. The waterline is at the S mark on the Plimsoll line. The ship travels to different ports.

Which option correctly shows how the ship will sit in the water at different ports?

	Ship travels to a	Ship will
(A)	tropical freshwater river port.	float higher in the water.
(B)	cool freshwater river port.	float higher in the water.
(C)	tropical sea port.	sink lower in the water.
(D)	cool water sea port.	sink lower in the water.

32. Contour lines on a map join places that are the same height above sea level.

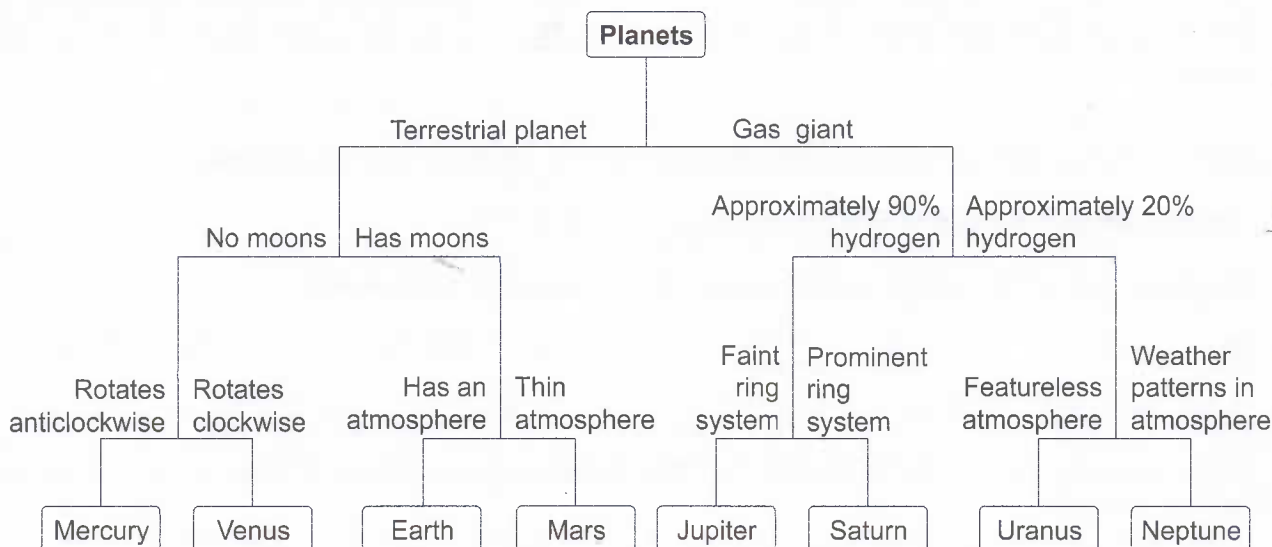
The map shows the contour lines for an area with a road passing through it. There are two towers, each ten metres high. One of the towers is located at X and the other at Y.

From which place along the road is only one of the towers visible?



33. Dichotomous keys are used to identify objects or to classify them into groups.

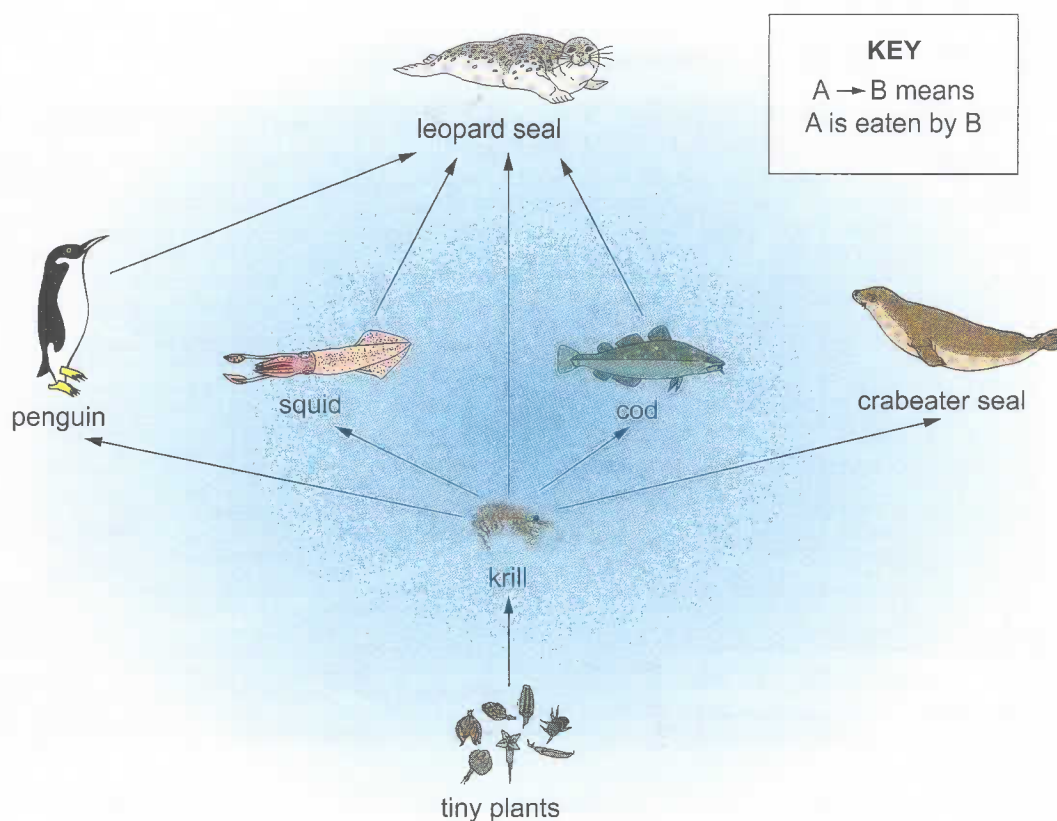
This dichotomous key can be used to identify the planets of the Solar System.



Using this dichotomous key, what is the correct description of Mars?

- (A) a terrestrial planet with moons and thin atmosphere
 (B) a terrestrial planet with no moons and thin atmosphere
 (C) a terrestrial planet with moons and rotates clockwise
 (D) a terrestrial planet with no moons and rotates clockwise

34. A food web is a diagram that shows the transfer of energy as one species is eaten by another species. The diagram represents part of the food web in a marine ecosystem.



What would be the likely result of a large decrease in the number of penguins?

- | | |
|--|---|
| (A) a decrease in the cod population | (B) an increase in the leopard seal population |
| (C) an increase in the number of tiny plants | (D) a decrease in the crabeater seal population |

35. At 20 °C, the largest amount of table salt you can dissolve in 100 mL of water is 36 grams.

A student added 144 grams of table salt to 300 mL of water at 20 °C.

How much more water does the student need to add to dissolve all the table salt?

- | | | | |
|------------|------------|------------|------------|
| (A) 100 mL | (B) 150 mL | (C) 400 mL | (D) 750 mL |
|------------|------------|------------|------------|

36. At sea level water boils at 100 °C. At higher altitudes, the atmospheric pressure is lower and water boils at a lower temperature.

It takes 3 minutes to cook an egg in boiling water at sea level.

How long will it take to cook an identical egg in boiling water at 8000 metres above sea level?

- | |
|-----------------------------|
| (A) 3 minutes |
| (B) more than 3 minutes |
| (C) less than 3 minutes |
| (D) the water does not boil |

37. Different substances have different densities.

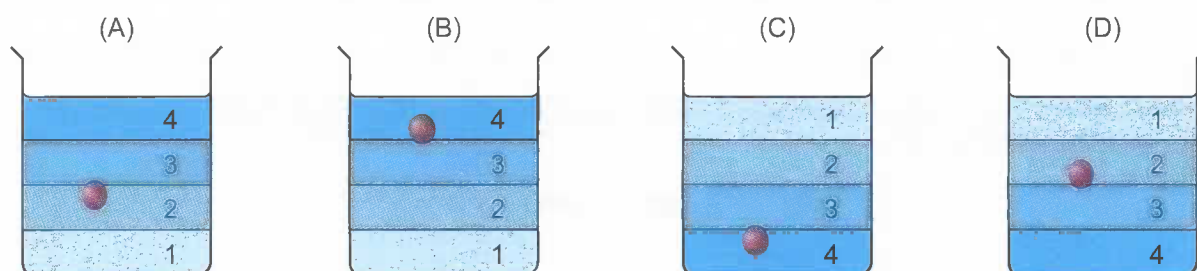
A less dense liquid will float on a liquid that is more dense. A solid will sink in a liquid that is less dense than the solid, and float on a liquid that is more dense.

Ben had four liquids with different densities. The table shows the liquids from least dense to most dense.

	Least dense			Most dense
Liquid	1	2	3	4

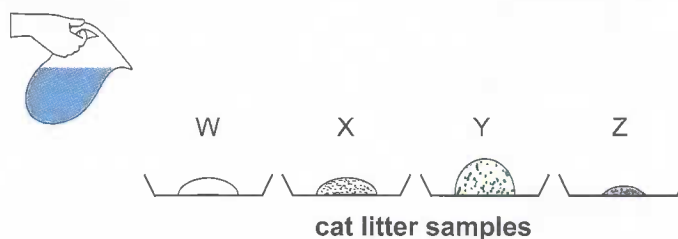
Ben dropped a grape into a beaker containing a layer of each of these liquids. The grape is less dense than liquid 3.

Which beaker shows the result of Ben's experiment?



38. Cat litter is made of materials that absorb liquids.

Zac wanted to find out which of four brands of cat litter is best for absorbing liquids. He measured the maximum amount of water absorbed by each sample of four different brands of cat litter.



The table summarises the observations Zac made during his experiment.

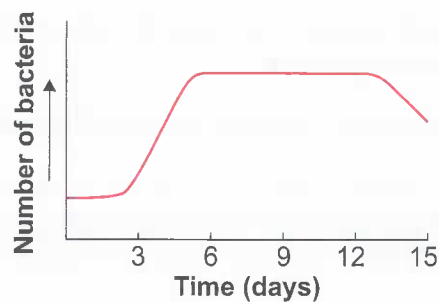
Brand of cat litter	Maximum amount of water absorbed by cat litter (cups)	Average size of crystals in cat litter (mm)	Colour of cat litter
W	2	2	colourless
X	2	1	white
Y	3	3	white and green
Z	$\frac{1}{2}$	1.5	white and blue

Zac's experiment is not a fair test.

Which of these is the reason that Zac's experiment is not a fair test?

- (A) Each sample of cat litter absorbed different amounts of water.
- (B) The amount of cat litter used in the trial was not measured.
- (C) The average size of the crystals in the cat litter was different.
- (D) He did not use the same colour of cat litter.

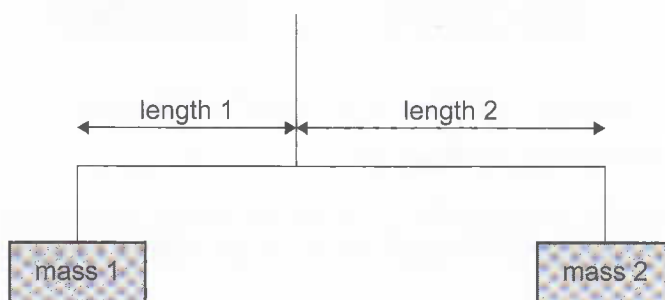
39. The graph shows the change in the number of bacteria living on a slice of stale bread over a given time.



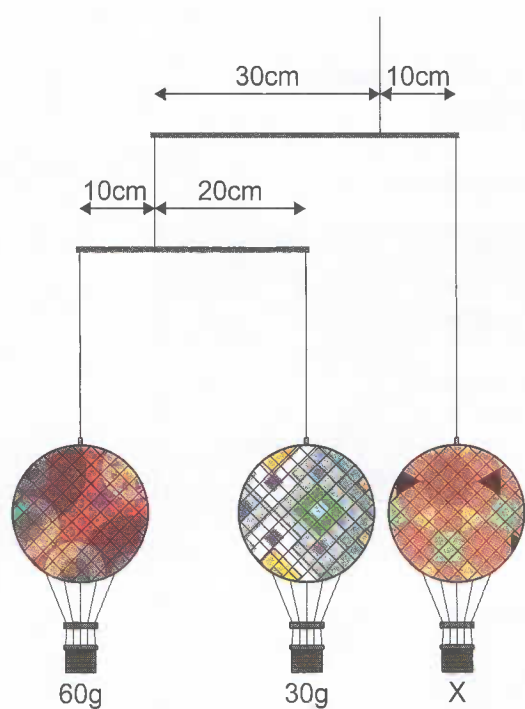
On which days was the number of bacteria that died greater than the number of bacteria that formed?

- (A) days 1–3 (B) days 3–6 (C) days 6–12 (D) days 13–15
40. Isabel made a mobile to hang in her room. To make sure that each section of her mobile was balanced, Isabel used the following rule:

$$\text{mass 1} \times \text{length 1} = \text{mass 2} \times \text{length 2}$$



The diagram shows the mobile she constructed.



What is the mass of balloon X?

- (A) 30 g
(B) 90 g
(C) 180 g
(D) 270 g

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Acknowledgment

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The following year levels should sit **THIS Paper**:

Australia	Year 5
Brunei	Primary 5
Indonesia	Year 6
Malaysia	Standard 5
New Zealand	Year 6
Pacific Region	Year 6
Singapore	Primary 4
South Africa	Grade 5



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ANALYSIS BY QUESTION

The following table shows the questions you answered correctly ☒ and shades the questions that you answered incorrectly ☐. The questions are ranked from hardest at the top to easiest at the bottom.

	Question content	Area assessed	Question number	Correct answer	Your answer	Students correct in AU
Difficult Questions	Calculate the mass required to make a mobile balance	Reasoning/Problem solving	40	D		9%
	Predict the effect of change on a food web	Predicting/Concluding	34	A	D	14%
	Estimate the time difference between two cities using their longitudes	Observing/Measuring	27	A		18%
	Calculate the distance to a storm using the speed of sound	Reasoning/Problem solving	10	A		19%
	Determine the length of a Blue Whale	Observing/Measuring	25	A	✓	30%
	Interpret Plimsoll lines	Observing/Measuring	31	C		32%
	Determine the solubility of salt	Observing/Measuring	35	A	✓	32%
	Determine visibility using contour lines	Reasoning/Problem solving	32	B	D	34%
	Recognise the relationship between altitude and boiling point	Observing/Measuring	36	B	A	35%
	Identify an appropriate control for an investigation	Investigating	12	B	✓	36%
	Use a column graph to convert wind speed from knots to km/h	Interpreting	19	B	✓	38%
	Interpret graphical data	Observing/Measuring	16	B	✓	39%
	Recognise why a test is not valid	Investigating	38	B	C	39%
	Calculate the volume of an object by displacement of water	Observing/Measuring	29	D	✓	41%
	Match diagram to show relative density of materials	Reasoning/Problem solving	37	D	B	41%
	Use a reading on a set of scales to determine the mass of a jug	Observing/Measuring	15	A	✓	42%
	Identify the section of a graph where death rate was higher than birth rate	Interpreting	39	D		42%
	Extract data from a graphical representation	Interpreting	7	D	A	45%
	Measure the volume of an object using displacement	Observing/Measuring	4	C	✓	50%
	Locate a fossil using map coordinates	Predicting/Concluding	26	A	✓	51%
	Recognise the use of the Plimsoll lines	Observing/Measuring	30	D		54%
	Determine the direction and relative speed of a gear	Reasoning/Problem solving	13	C	✓	55%
	Conclude that diffusion and osmosis have occurred	Predicting/Concluding	24	C		58%
	Determine the highest daily wind speed from a column graph	Interpreting	18	A		61%
	Interpret sector bar graphs	Interpreting	22	D	✓	63%
	Sequence the relative time of different footprints	Reasoning/Problem solving	23	B	✓	64%
Easy Questions	Identify the most concentrated solution	Predicting/Concluding	14	B	D	65%
	Recognise the effect of air resistance on objects with a different surface area	Predicting/Concluding	9	C	✓	67%
	Predict the time for an event to occur	Investigating	1	B	C	67%
	Recognise which magnets will attract or repel	Reasoning/Problem solving	28	B	✓	68%
	Identify best method for separating iron filings	Investigating	8	C	D	68%
	Infer the distance to an object using echo timing	Predicting/Concluding	2	B	✓	71%
	Use a dichotomous key to identify planet Mars	Predicting/Concluding	33	A	✓	75%
	Identify a graph to match given data	Observing/Measuring	3	A	✓	80%
	Sequence stages in the carbon cycle	Observing/Measuring	17	A	✓	82%
	Determine the order of a series of photos of a flowering cactus plant	Observing/Measuring	5	C	✓	82%
	Rank insulating materials based on their effectiveness	Predicting/Concluding	11	A	✓	83%
	Match relevant data to a graphical representation	Interpreting	6	D	✓	84%
	Determine the percentage of sulfur in sulfuric acid	Reasoning/Problem solving	20	C	✓	84%
	Interpret sector bar graphs	Observing/Measuring	21	A	✓	85%

Access to Assessment Questions

This question analysis is most helpful when used in conjunction with examples of the questions. Student reports and questions are available from the Student Portal at www.eaa.unsw.edu.au/student-portal. You will need your TAP-ID (0715-9907-27) and your PIN (2497) to login.